Hypothesis Testing I

Unit - 4

Session - 8

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Agenda

- Test of proportion
  - One sample (Z proportion test)
  - 2 sample (Z proportion test)
- Non Parametric tests
  - Spearman Rank Correlation
  - Wilcoxon signed rank test
  - Manwhitney U test

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Test for Population Proportion

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Test for proportion

- For qualitative data the proportion of a desired characteristic is obtained
- Test for proportion:
  - One sample: Testing population proportion (P) is equal to a specified value (P_0)
  - Two sample: Testing equality of Two population proportions ($P_1 = P_2$)
- Similar to the tests of population mean

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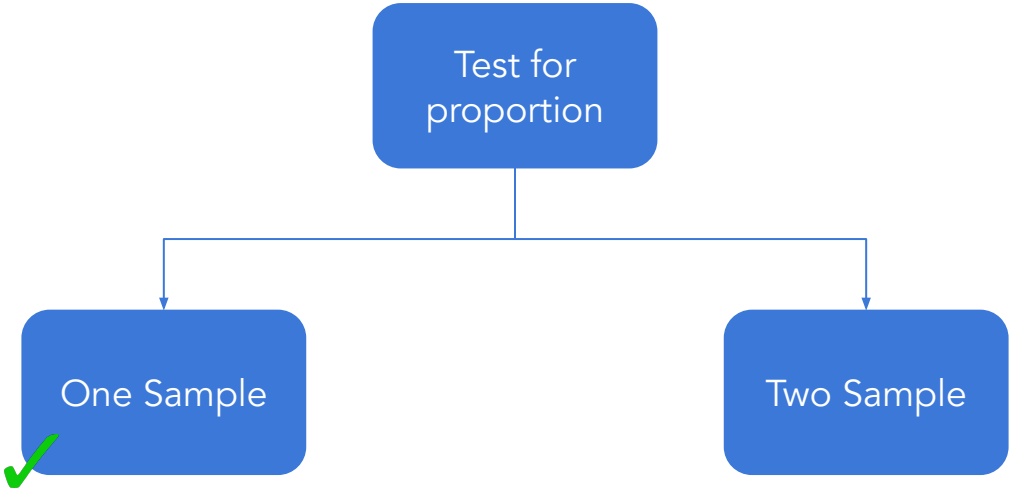
Test for proportion



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One sample test - hypothesis

- The hypothesis to test the population proportion is equal to a specified value

$$H_0 : P = P_0 \quad \text{against} \quad H_1 : P \neq P_0$$

- It implies

H_0: The population proportion is equal to P_0

against H_1: The population proportion is not equal to P_0

- Failing to reject H_0 implies that the population proportion is equal to P_0

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Test for proportions

- The test statistic is given by

Sample proportion

Specified proportion

$$Z = \frac{p - P_0}{\sqrt{\frac{P_0(1-P_0)}{n}}}$$

Sample size

- Under H_0, the test statistic follows standard normal distribution

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One sample test for proportion - decision rule

	H_1	Based on critical region	Based on p-value	Based on confidence interval
For two tailed test	$P \neq P_0$	Reject H_0 if $\lvert Z \rvert \geq Z_{\alpha/2}$	Reject H_0 if p-value is less than or equal to the level of significance	Reject H_0 if P_0 does not lie in the confidence interval
For left tailed test	$P < P_0$	Reject H_0 if $Z \leq -Z_\alpha$		
For right tailed test	$P > P_0$	Reject H_0 if $Z \geq Z_\alpha$		

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One sample test for proportion

Question:

From a sample 361 business owners had gone into bankruptcy due to recession. On taking a survey, it was found that 105 of them had not consulted any professional for managing their finance before opening the business. Test the null hypothesis that at most 25% of all businesses had not consulted before opening the business.

Test the claim using p-value technique. [Use $\alpha = 0.05$].

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One sample test for proportion

Solution:

From a sample 361 business owners had gone into bankruptcy due to recession.
i.e. $n = 361$

On taking a survey, it was found that 105 of them had not consulted any professional for managing their finance before opening the business.

Let X: business which did not consult before
$x = 105$

The sample proportion $(p) = x/n = 105/361 = 0.2909$

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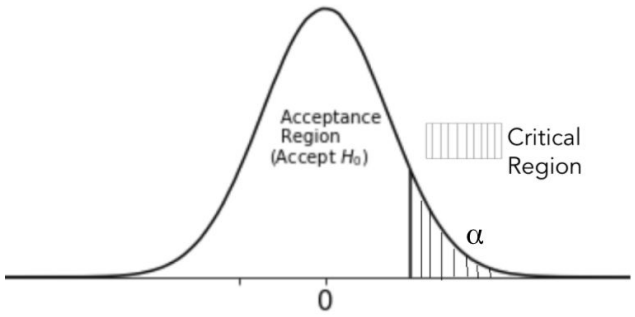
One sample test for proportion

Solution:

To test: The null hypothesis that at most 25% of all businesses had not consulted before opening the business

Here $P_0 = 0.25$

To test, H_0: $P \leq 0.25$ against H_1: $P > 0.25$



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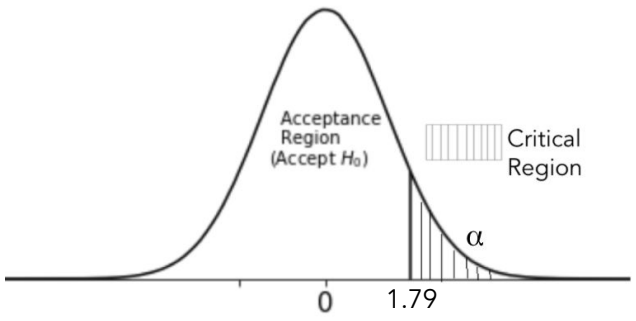
One sample test for proportion

Solution:

The test statistic $Z = \frac{p - P_0}{\sqrt{\frac{P_0(1-P_0)}{n}}} = \frac{0.2909 - 0.25}{\sqrt{\frac{0.25(0.75)}{316}}} = 1.79$



The p-value = P(Z > 1.79) = 0.0367

As p-value < 0.05, reject H_0.

We may conclude that at least 25% of all businesses had not consulted before starting the business.

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One sample test for proportion

Question:

Suppose you work for a company that produces a new brand of energy drink. In a previous study, 60% of consumers said they would recommend the drink to their friends. You decide to conduct a new survey to check if the proportion of users willing to recommend the drink has changed. You survey 150 consumers, and 90 of them say they would recommend it.

You want to test the hypothesis at a significance level of $\alpha=0.05$.

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- Null Hypothesis (H_0): The proportion of consumers who would recommend the drink remains the same, $p=0.60$.
- Alternative Hypothesis (H_a): The proportion of consumers who would recommend the drink has changed, $p \neq 0.60$.

- Sample size (n) = 150
- $p=90/150 = 0.60$

$$Z = \frac{p - P_0}{\sqrt{\frac{P_0(1-P_0)}{n}}}$$

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$$z = \frac{0.60 - 0.60}{\sqrt{\frac{0.60(1-0.60)}{150}}}$$

$$z = \frac{0.60 - 0.60}{0.04} = \frac{0}{0.04} = 0$$

critical z-value for α=0.05 = 1.96

- $Z_{STAT} < Z_{CRIT}$ =0<1.96 , we fail to reject the null hypothesis.

we conclude that there is not enough evidence to suggest that the proportion of consumers who would recommend the drink has changed from the previous proportion of 60%.

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Test for proportion

When the outcome is discrete or dichotomous (success or failure), the test objective is :

Compare the proportion of successes (P-hat or P) in a single population to a known proportion (P_0)

This test will assess whether or not a sample from a population represents the true proportion from the entire population.

The proportion P-hat is calculated using the ratio of no. of successes to the sample size (n)

P-hat = x/n

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Standard Deviation and Z score for Test of proportion

- Standard deviation (σ) of the sampling distribution

$$se = sqrt[P_0 * (1 - P_0) / n]$$

where P_0 is the hypothesized value of population proportion in the null hypothesis, and n is the sample size.

- Test statistic. The test statistic is a z-score (z) defined by the following equation.

$$z = (p - P_0) / se$$

where P_0 is the hypothesized value of population proportion in the null hypothesis, P is the sample proportion (sometimes referred to as P-hat), and σ is the standard deviation of the sampling distribution.

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Test of proportion with One Sample - Critical Value Approach

1. State the null hypothesis H_o and the alternative hypothesis H_A.
2. Calculate the test statistic:

$$z = \frac{\hat{p} - p_0}{\sqrt{\frac{p_0(1-p_0)}{n}}}$$

where p_0 is the null hypothesized proportion i.e., when H_o: $P=P_0$

P-hat is the sample proportion

3. Determine the critical region.
4. Make a decision. Determine if the test statistic falls in the critical region. If it does, reject the null hypothesis. If it does not, do not reject the null hypothesis.

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Test of proportion with One Sample - p-Value Approach

1. State the null hypothesis H_o and the alternative hypothesis H_A.
2. Calculate the test statistic:

$$z = \frac{\hat{p} - p_0}{\sqrt{\frac{p_0(1-p_0)}{n}}}$$

where p0 is the null hypothesized proportion i.e., when H_o: $P=P_0$

P-hat is the sample proportion

3. Calculate the p-value.
4. Make a decision. Check whether to reject the null hypothesis by comparing p-value to α. If the p-value < α, then reject Ho; otherwise do not reject Ho.

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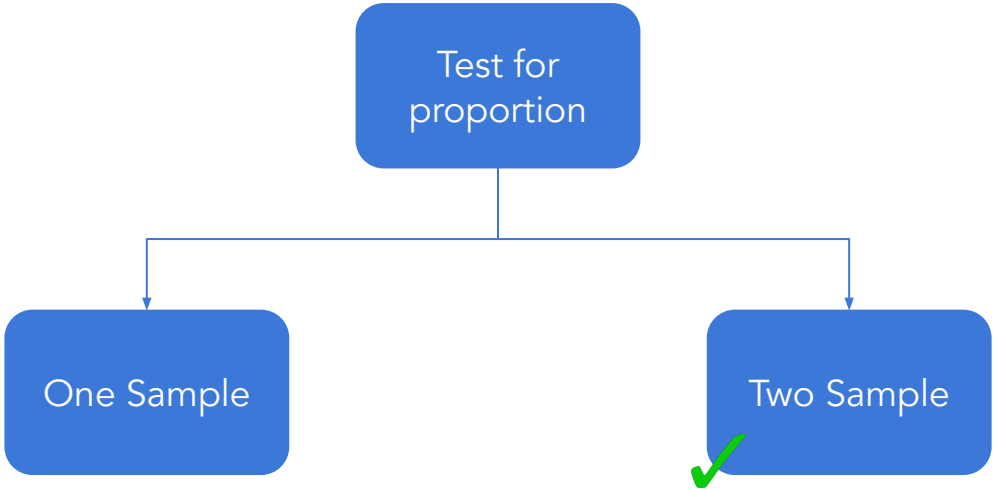
Test for proportion



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Test of proportion with Two Independent Samples

Objective:

Compare proportions of successes between the two groups. The relevant sample data are the sample sizes in each comparison group (n_1 and n_2) and the sample proportions (P_1-hat and P_2-hat) which are computed by taking the ratios of the numbers of successes (x_1 and x_2) to the sample sizes in each group, i.e.,

$$\hat{p}_2 = \frac{x_2}{n_2} \quad \text{and} \quad \hat{p}_1 = \frac{x_1}{n_1}$$

Test Statistic : $$z = \frac{\hat{p}_1 - \hat{p}_2}{\sqrt{\hat{p}(1-\hat{p})\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}$$ where $\hat{p} = \frac{x_1 + x_2}{n_1 + n_2}$

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Two sample tests for population proportion

- Let there be two samples sizes n_1 and n_2 from different populations of such that x_1 and x_2 are the number of specific items in each of them respectively

- Suppose these samples have proportions of specific items p_1 and p_2 respectively

- To test the equality of population proportion from which these samples are chosen

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Two sample test - hypothesis

- The hypothesis to test the population proportion

$$H_0 : P_1 = P_2 \quad \text{against} \quad H_1 : P_1 \neq P_2$$

- It implies

H_0: The two population proportions are equal ($P_1 = P_2$)

against H_1: The two population proportions are not equal ($P_1 \neq P_2$)

- Failing to reject H_0 implies that the two population proportions are equal

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Test for proportions

- The test statistic is given by

$$Z = \frac{p_1 - p_2}{\sqrt{\bar{P}(1-\bar{P})(\frac{1}{n_1} + \frac{1}{n_2})}}$$

where $\bar{P}$ is the proportion of pooled sample such that

$$\bar{P} = \frac{x_1 + x_2}{n_1 + n_2} = \frac{n_1 p_1 + n_2 p_2}{n_1 + n_2}$$

- Under H_0, the test statistic follows standard normal distribution

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Two sample test for proportion - decision rule

	H_1	Based on critical region	Based on p-value	Based on confidence interval
For two tailed test	$P_1 \neq P_2$	Reject H_0 if $\lvert Z \rvert \geq Z_{\alpha/2}$	Reject H_0 if p-value is less than or equal to the level of significance	Reject H_0 if $P_1 - P_2$ does not lie in the confidence interval
For left tailed test	$P_1 < P_2$	Reject H_0 if $Z \leq -Z_\alpha$		
For right tailed test	$P_1 > P_2$	Reject H_0 if $Z \geq Z_\alpha$		

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Two sample test for proportion

Question:

Steve owns a kiosk where sells two magazines - A and B in a month. He buys 100 copies of magazine A out of which 78 were sold and 70 copies of magazine B out of which 65 were sold. Is there enough evidence to say that magazine is B is more popular?

Test the claim using p-value technique. [Use $\alpha = 0.05$].

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Two sample test for proportion

Solution:

Steve owns a kiosk where sells two magazines - A and B in a month.

Let X: the number of magazines sold

Out of 100 copies of magazine A 78 are sold
Here, $x_1 = 78$ and $n_1 = 100$
Let p_1 be the proportion of sell of magazine A
$p_1 = x_1/n_1 = 78/100 = 0.78$

Out of 70 copies of magazine B 65 are sold
Here, $x_2 = 65$ and $n_1 = 70$
Let p_2 be the proportion of sell of magazine B
$p_2 = x_2/n_2 = 65/70 = 0.928$

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Two sample test for proportion

Solution:

To test, whether magazine B is more popular, i.e

$H_0: P_1 \geq P_2$ against $H_1: P_1 < P_2$

Where P_1: denotes population proportion of magazine A sold
P_2: denotes population proportion of magazine B sold

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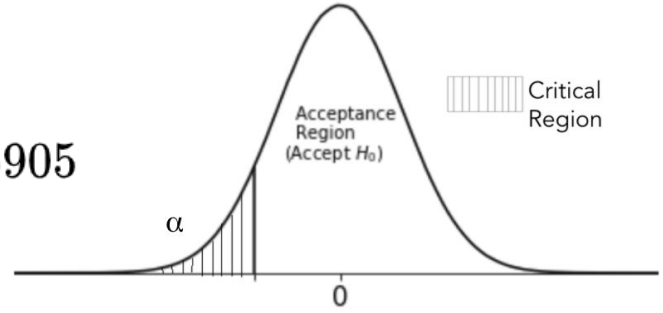
Two sample test for proportion

Solution:

The pooled proportion is $\bar{P} = \frac{x_1+x_2}{n_1+n_2} = \frac{78+65}{100+70} = 0.84$

The test statistic is

$$Z = \frac{p_1 - p_2}{\sqrt{\bar{P}(1-\bar{P})(\frac{1}{n_1}+\frac{1}{n_2})}} = \frac{0.78-0.928}{\sqrt{0.84(1-0.84)(\frac{1}{100}+\frac{1}{70})}} = -2.5905$$



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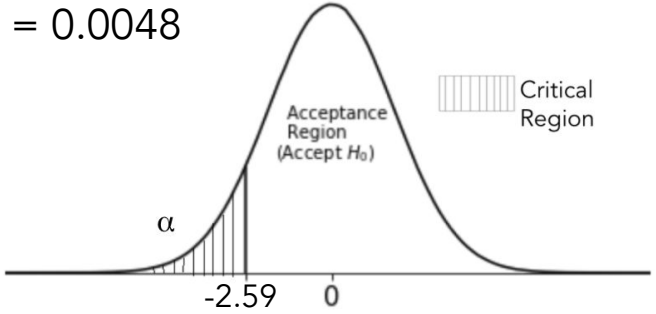
Two sample test for proportion

Solution:

The test statistic Z = -2.5905

The p-value = P(Z < Z_{calc}, under H_0) = P(Z < -2.5905, μ = 13) = 0.0048

Since p-value < 0.05, we reject H_0.



Thus there is enough evidence to conclude that magazine is B is more popular.

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Two sample test for proportion

A marketing team wants to compare the effectiveness of two different advertising campaigns (Campaign A and Campaign B) in terms of the proportion of customers who makes a purchase. They run the two campaigns independently and gather the following data:

Campaign A: Out of 400 customers, 120 made a purchase.

Campaign B: Out of 500 customers, 200 made a purchase.

You want to test if there is a significant difference in the purchasing proportions between the two campaigns at a significance level of $\alpha=0.1$.

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- Null Hypothesis (H_0): The proportions are equal, $p_1 = p_2$.
- Alternative Hypothesis (H_a): The proportions are not equal, $p_1 \neq p_2$.

For Campaign A:

- Sample size (n_1) = 400
- Number of successes (purchases) = 120
- Sample proportion (p_1) $= 120/400 = 0.30$

For Campaign B:

- Sample size (n_2) = 500
- Number of successes (purchases) = 200
- Sample proportion (p_2) $= 200/500 = 0.4$

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$$\bar{P} = \frac{x_1+x_2}{n_1+n_2} = \frac{120+200}{400+500} = \frac{320}{900} \approx 0.3556$$

$$Z = \frac{p_1 - p_2}{\sqrt{\bar{P}(1-\bar{P})(\frac{1}{n_1}+\frac{1}{n_2})}}$$

$$z = \frac{0.30 - 0.40}{0.0313} = \frac{-0.10}{0.0313} \approx -3.19$$

critical z-value for α=0.1 = 1.645

- $Z_{STAT} > Z_{CRIT}$ =3.19>1.645 , we reject the null hypothesis.

we conclude that there is a significant difference in the proportion of customers who made a purchase between Campaign A and Campaign B. Specifically, Campaign B appears to be more effective than Campaign A, given that a higher proportion of customers who saw Campaign B made a purchase..

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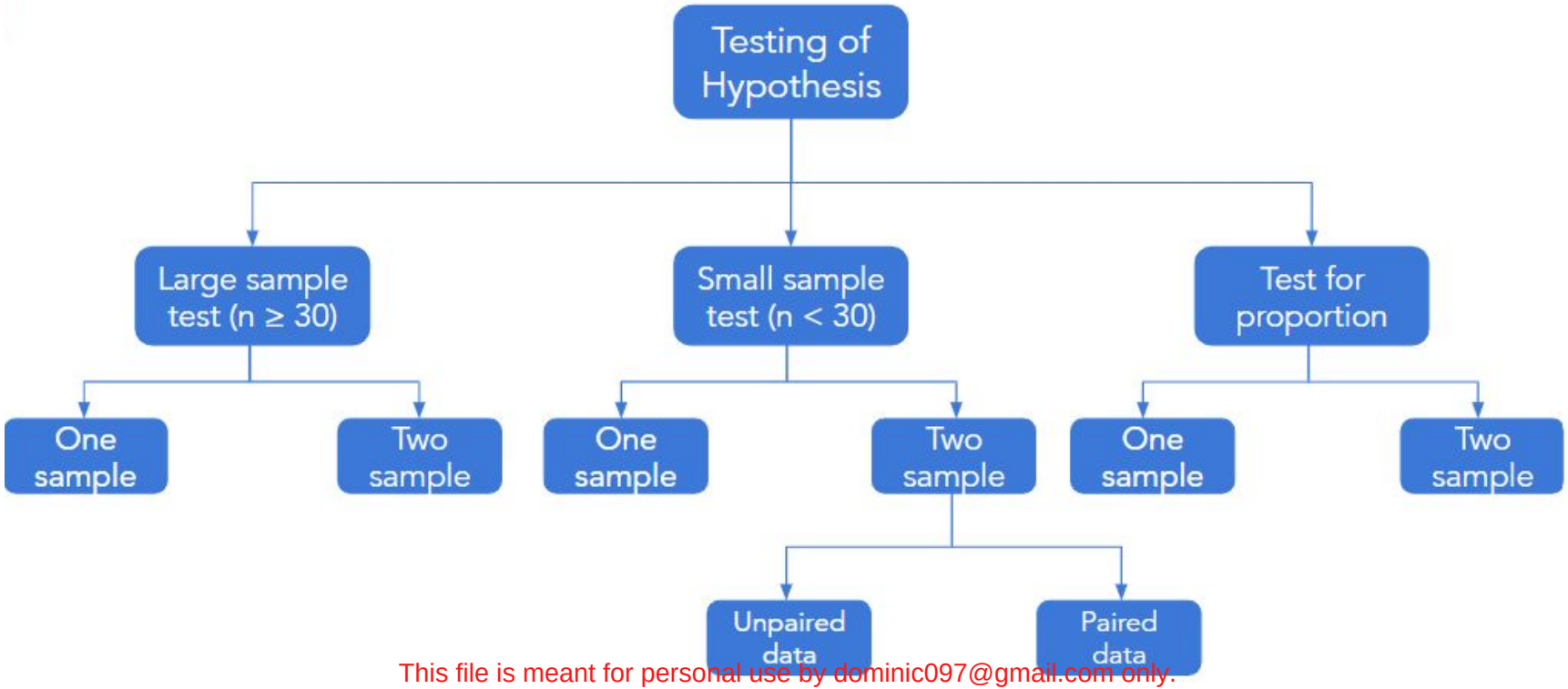
Summary



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Parametric & Non parametric tests

- The tests considered so far have two features:
  - The probability distribution of the samples was assumed to be known
  - The hypothesis test was about the parameter of the probability distribution
- These tests are known as the parametric tests
- The times when these assumptions are not satisfied, use the non-parametric tests

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Parametric & Non parametric tests

Parametric tests	Non-parametric tests
It makes assumptions about the parameters of the population distribution(s) from which one's data are drawn	It makes no such assumptions
The information about the population is completely known by means of its parameters	There is no information about the population but still it is required to test the hypothesis of the population,
The data should be normally distributed	Data does not follow any specific distribution, So, it is known as ***"distribution free tests"***
Null hypothesis is made on parameters of the population distribution	The null hypothesis is free from parameters
It is applicable only for variable	It is applicable for both variables and attributes
No parametric tests for Nominal scale data	It exists for nominal and ordinal scale data
It is most powerful	It is not as powerful as parametric test

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Non- Parametric Test

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Common Non-Parametric tests

Parametric Test		Non-parametric Test
Association of relationship- Karl pearson correlation	→	Spearman's rank correlation
Between Subjects t-test	→	Mann-Whitney *U* Test
Within Subjects t-test	→	Wilcoxon Signed Ranked Test

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Spearman rank correlation

The Spearman's rank correlation test is a measure of association based on the ordinal features of data.

This method is simple to use and easy to apply.

The Spearman's coefficient of rank correlation 'ρ' is given by,

$$\rho = 1 - \frac{6\sum_{i=1}^{n} d_i^2}{n^3 - n}$$

where,

'd' is the difference between the two ranks and 'n' is the number of ranks.

The limits for rank correlation coefficient are $-1 \leq \rho \leq 1$.

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Question

Consider a study that examines the relationship between the number of hours students study and their scores on a test. Here are the data for 8 students:

Student	Hours Studied	Test Score
1	4	85
2	2	70
3	3	80
4	5	88
5	6	90
6	1	65
7	7	92
8	8	95

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Soln:

Student	Hours Studied	Rank (Hours)	Test Score	Rank (Score)
1	4	4	85	4
2	2	2	70	2
3	3	3	80	3
4	5	5	88	5
5	6	6	90	6
6	1	1	65	1
7	7	7	92	7
8	8	8	95	8

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Student	Rank (Hours)	Rank (Score)	d=Rank (Hours)−Rank (Score)	d2
1	4	4	0	0
2	2	2	0	0
3	3	3	0	0
4	5	5	0	0
5	6	6	0	0
6	1	1	0	0
7	7	7	0	0
8	8	8	0	0

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$$\rho = 1 - \frac{6\sum_{i=1}^{n} d_i^2}{n^3 - n}$$

$$\rho = 1 - \frac{6 \times 0}{8(8^2 - 1)} = 1 - 0 = 1$$

The Spearman rank correlation coefficient ($\rho=1$) indicates a perfect positive correlation between the hours studied and the test scores, meaning that as the number of hours studied increases, the test scores also increase in a perfectly linear fashion.

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A manager wants to determine if there is a relationship between an employee's score on a creativity test (X) and their performance rating in their annual review (Y).

Employee	Creativity Score (X)	Performance Rating (Y)
1	95	80
2	80	75
3	70	60
4	90	95
5	75	70

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Soln

Employee	RankX	RankY	di=(RankX−RankY)	di2
1	5	4	5 - 4 = 1	1
2	3	3	3 - 3 = 0	0
3	1	1	1 - 1 = 0	0
4	4	5	4 - 5 = -1	1
5	2	2	2 - 2 = 0	0
			Total	**2**

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$$\rho = 1 - \frac{6\sum_{i=1}^{n} d_i^2}{n^3 - n}$$

$$\rho = 1 - \frac{6 \times 2}{5(5^2 - 1)}$$

= 0.9

The Spearman rank correlation coefficient ($\rho = 0.9$) This indicates a very **strong, positive, monotonic relationship** between the creativity score and the performance rating. In simple terms, employees who ranked high on the creativity test also tended to rank high in their annual performance review.

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Wilcoxon Signed Rank test

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Wilcoxon signed rank test

- The *Wilcoxon* signed-rank test is proposed by *Frank Wilcoxon* (1945) and it does not take into consideration the magnitude of the difference between two equal groups
- It is more powerful than the sign test because it tests not only direction but also the magnitude of differences within pairs of matched groups.
- This test like the sign test deals with dependent groups made up of matched pairs of individuals and is not applicable to independent groups.

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Wilcoxon signed rank test

- The Wilcoxon signed-rank test applies in the case of a symmetric continuous distribution. Under this condition, we can test the null hypothesis (H0: $\mu 1 = \mu 2$).
- Rank the differences of the paired observations without regard to sign and assign the ranks.
- Calculate the totals for number of +ve signs and –ve signs and take the consideration into the account for smallest totals of signed values. The critical value of w can be obtained from table.
- Test the null hypothesis, i.e., the null hypothesis (H0) is rejected if the computed value of lowest signed value is less than or equal to (<=) the appropriate table value at the level of significance (say, 0.05).
- i.e., The criteria to accept (or) reject the null hypothesis is, if W ≤ critical value then reject H0, otherwise Accept H0.

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Question

A new energy drink company claims their product will improve a person's endurance score. An independent researcher tests 6 volunteers *before* and *after* drinking the product. The researcher wants to know if the difference is **significant**, without assuming the scores are normally distributed.

Volunteer	Before Score (X)	After Score (Y)
1	22	25
2	15	18
3	10	11
4	28	27
5	14	20
6	17	17

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Soln:

Volunteer	Before (X)	After (Y)	Difference (di)	Sign	Action
1	22	25	3	+'	Keep
2	15	18	3	+'	Keep
3	10	11	1	+'	Keep
4	28	27	-1	-'	Keep
5	14	20	6	+'	Keep
6	17	17	0	N/A	**Exclude**

look at the absolute values of the remaining differences: 3, 3, 1, 1, 6

- |d_i|=1 appears twice, occupying ranks 1 and 2. **Tie Rank:** (1+2)/2 = 1.5
- |d_i|=3 appears twice, occupying ranks 3 and 4. **Tie Rank:** (3+4)/2 = 3.5
- |d_i|=5 is the largest, getting rank **5**.

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Volunteer	Sign	Rank	Signed Rank
1	+'	3.5	3.5
2	+'	3.5	3.5
3	+'	1.5	1.5
4	-'	1.5	-1.5
5	+'	5	5

Sum of Positive Ranks (W^+): 3.5 + 3.5 + 1.5 + 5.0 = 13.5

Sum of Negative Ranks (W^-): 1.5 = 1.5

The test statistic (W) is the **smaller** of the two sums.

W = min(13.5, 1.5) = 1.5

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- **Sample Size (n):** 5
- **Significance Level α :** 0.05 (two tailed)

Using the Wilcoxon Signed-Rank Critical Values Table for n=5 and alpha=0.05 (two-tailed), the critical value (W_crit) is **0**.

W >W_crit (Fail to Reject the Null Hypothesis.)

While four out of five people improved, the overall strength of the change was not different enough from zero to be considered a significant effect at the 0.05 level. The tiny amount of negative change (Volunteer 4) was enough to keep the result non-significant, suggesting the energy drink's effect might have been due to chance.

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Table

Question

A manager wants to know if the median daily call volume of her team is different from the company's target of **100 calls**.

Employee	Average Daily Calls (xi)
A	95
B	108
C	120
D	90
E	102
F	88
G	115

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Soln:

Employee	Score (xi)	di=xi−100	Sign
A	95	-5	-'
B	108	8	+'
C	120	20	+'
D	90	-10	-'
E	102	2	+'
F	88	-12	-'
G	115	15	+'

Rank the Absolute Differences

Employee	Difference (di)	Rank	Signed Rank
A	-5	2	**-2**
B	8	3	**3**
C	20	7	**7**
D	-10	4	**-4**
E	2	1	**1**
F	-12	5	**-5**
G	15	6	**6**

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Sum the Signed Ranks to Find the Test Statistic (W)

- **Sum of Positive Ranks (W^+):** 3 + 7 + 1 + 6 = 17
- **Sum of Negative Ranks (W^-):** 2 + 4 + 5 = 11

The **calculated test statistic (W)** is the **smaller** of the two sums.

W = min(17, 11) = 11

Calculated W: 11
Critical W (for n=7, alpha=0.05 two-tailed): 2

W >W_crit (Fail to Reject the Null Hypothesis.)

Since 11 is **not less than or equal to** 2, we **Fail to Reject the Null Hypothesis (H0)**.

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Man whitney - U Test

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Mann-Whitney U test

- This non-parametric test for two samples was described by *Wilcoxon* and studied by *Mann* and *Whitney*.
- The Mann-Whitney U test is used to compare difference between two independent groups when the dependent variable is either ordinal or continuous, but not normally distributed.
- It is most widely used test as alternative to independent sample t-test an (parametric).
- It works on *interval* and *ordinal* data.
- It can be used for very small samples.

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Question

A tech company wants to compare the efficiency of two different coding teams, Team Alpha and Team Beta, based on the time (in hours) it took each team to complete a standardized project. The sample sizes are small, and the data is likely not normally distributed.

Team Alpha (n1=5)	Team Beta (n2=6)
15	18
12	19
17	25
10	20
14	22
	16

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Soln:

Null Hypothesis (H0): The median completion times for Team Alpha and Team Beta are the same $\mu 1 = \mu 2$.

Alternative Hypothesis (H1): The median completion times are different $\mu 1 \neq \mu 2$, (Two-Tailed Test).

Significance (alpha): 0.05

combine all N = 5 + 6 = 11 scores and assign a rank (1 being the fastest time, 11 being the slowest).

Time (Hours)	Team	Combined Rank
10	Alpha	1
12	Alpha	2
14	Alpha	3
15	Alpha	4
16	Beta	5
17	Alpha	6
18	Beta	7
19	Beta	8
20	Beta	9
22	Beta	10
25	Beta	11

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separate the ranks back into their original teams and sum them up (R1 and R2).

Team Alpha Ranks (R1)	Team Beta Ranks (R2)
1	5
2	7
3	8
4	9
6	10
sum (R1) =16	11
	sum (R2) = 50

$$U_1 = n_1 n_2 + \frac{n_1(n_1+1)}{2} - R_1$$

$$U_2 = n_1 n_2 + \frac{n_2(n_2+1)}{2} - R_2$$

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For Team Alpha (U_1):

$$U_1 = (5)(6) + \frac{5(5+1)}{2} - 16$$

$$U_1 = 30 + \frac{30}{2} - 16 = 30 + 15 - 16 = \mathbf{29}$$

For Team Beta (U_2):

$$U_2 = (5)(6) + \frac{6(6+1)}{2} - 50$$

$$U_2 = 30 + \frac{42}{2} - 50 = 30 + 21 - 50 = \mathbf{1}$$

The calculated test statistic (U) is the smaller of the two values:

U = min(U1, U2) = min(29, 1) = 1

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n1=5, n2=6, and a two-tailed alpha=0.05.

- **Calculated U**: 1
- **Critical U Value (from table)**: 3

Decision Rule: Reject H0 if the calculated U is **less than or equal to** the critical U value.

Since 1<3 , we **Reject the Null Hypothesis (H0)**.

We conclude that there is a **statistically significant difference** in the project completion times between the two teams at the 0.05 level. Since Team Alpha had a much lower sum of ranks (R1=16) and a lower calculated U statistic (which means its ranks were consistently smaller), we conclude that **Team Alpha's median completion time is significantly faster than Team Beta's.**

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Table

Question

A company wants to see if there's a difference in customer satisfaction scores between two marketing campaigns.Consider it doesn’t follow normality

Campaign A (n1=5)	Campaign B (n2=6)
12	19
18	15
25	23
10	20
14	22
	16

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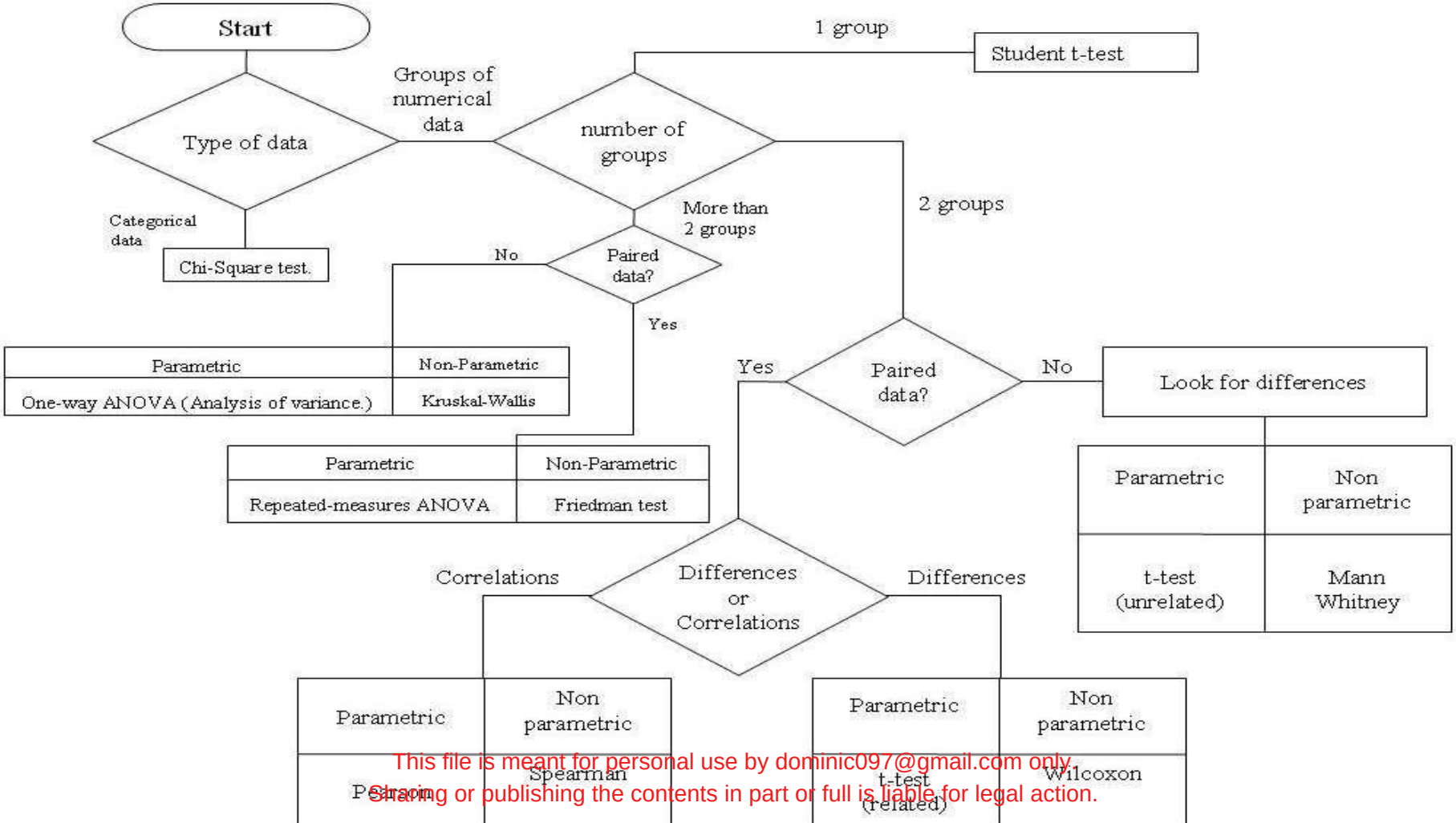

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Thank You

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